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Custom Optical Microscope and Devices

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[BISC6-01] Development of an Automated LEGO-based Inverted Fluorescence Microscope with Robust Autofocusing Capabilities

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We present a cost-efficient, automated LEGO-based inverted microscope optimized for fluorescence imaging. Featuring adaptive fluorescence autofocus and high spectral fidelity, this modular open-hardware platform functions as a reconfigurable optical breadboard for versatile biological research.

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Development of an Automated LEGO-based Inverted Fluorescence Microscope with Robust Autofocusing Capabilities

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ABSTRACT

Commercial inverted fluorescence microscopes are expensive and difficult to customize, limiting their use in both laboratory and field applications. This study introduces a cost-efficient (~$1300 USD) fully automated LEGO-based inverted fluorescence microscope that functions as a reconfigurable optical breadboard. Unlike conventional static frames, the modular architecture enables rapid modification of the optical train and component layout. A rigid gantry stage provides 0.625 μm positioning resolution for precise large-area scanning, operated through a user-friendly Web-based remote-control interface. A two-stage adaptive autofocus algorithm enables reliable brightfield and fluorescence focusing within 14 seconds, while a Color Correction Matrix (CCM) pipeline ensures high spectral fidelity. Ultimately, this "future-proof" open-hardware ecosystem enables users to easily upgrade or rearrange optical modules, offering a versatile platform for evolving biological imaging applications.

Keywords: LEGO-based Microscope, Inverted Microscope, Autofocusing, Modular Optics, Web-based Remote-control Interface

1. INTRODUCTION

Inverted fluorescence microscopy is essential for live-cell imaging but remains inaccessible due to the high cost and proprietary designs of commercial systems. Although open-source hardware initiatives have developed accessible alternatives, these solutions often come with specific mechanical compromises. For instance, the widely utilized OpenFlexure microscope [1] relies on flexure-based stages. While these offer frictionless precision, they are limited by short travel ranges due to material plasticity, rendering them unsuitable for scanning larger areas such as multi-well plates. Other low-cost designs predominantly use upright architectures, which are not ideal for adherent cells, or modular systems like UC2 [2] that suffer from mechanical and thermal stability issues arising from snap-fit connections. Additionally, these systems typically employ uncalibrated consumer-grade sensors with low sensitivity and high color crosstalk [3], which restricts their effectiveness for quantitative analysis. Furthermore, most DIY designs rely on monolithic 3D-printed frames, necessitating the redesign and reprinting of the entire instrument body when modifying its form factor.

To overcome these limitations, we introduce the concept of a "LEGO optical breadboard" for constructing a modular inverted fluorescence microscope that supports both automated large-area scanning and rapid optical prototyping. Unlike fixed commercial systems or monolithic DIY frames, our design leverages LEGO's inherent modularity to form a reconfigurable optical workbench, allowing users to freely adjust inter-component spacing, insert new optical modules, or reroute the light path as needed. Custom 3D-printed interfaces are integrated at critical nodes to ensure precise optical alignment. Mechanically, the restrictive flexure mechanism is replaced with a rigid gantry-style stage driven by lead screws, providing a significantly expanded working volume of 70 × 50 × 70 mm, and a theoretical positioning resolution of 0.625 μm through microstepping. With an estimated material cost of approximately USD 1,300, the system successfully combines the affordability of open-source hardware with the mechanical range, stability, and adaptability required for various biological imaging applications.

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2. MATERIALS AND METHODS

2.1. Hardware Architecture

The microscope chassis was constructed from high-precision LEGO bricks to function as a highly versatile optical breadboard. Unlike traditional fixed-body microscopes, this LEGO-based framework allows for the dynamic arrangement of optical components. The positions of the objective mount, filter cube, and optical lens can be easily shifted along the LEGO grid to accommodate different optical tube lengths or experimental geometries. Custom 3D-printed interfaces were incorporated at key structural nodes to reinforce stability and maintain concentricity along the optical axis, while allowing for rapid reconfiguration without the need for reprinting the entire frame. A 3D-printed sliding-filter drawer was integrated into the LEGO housing to enable quick fluorescence-filter exchange without disturbing the objective alignment.

The optical system is engineered to facilitate dual-modality imaging, employing a 10× magnification objective lens (A10, Olympus, Tokyo, Japan), providing sufficient magnification for cellular observation. The modular mounting interface is designed to enable the seamless interchange of the objective lens, thereby allowing for system upgrades to high-numerical-aperture (NA) or high-magnification optics to address a spectrum of resolution requirements. Brightfield illumination is sourced from a 6500 K white LED (XQEAWT-H0-0000-00000BFE1, Cree LED, Durham, NC), which is complemented by a condenser lens (ACL12708U, Thorlabs, Newton, NJ). For fluorescence excitation, a high-power blue LED (LZ4-40B208, LED Engin, Santa Clara, CA) is employed; this source is collimated by an aspheric condenser (15-194, Edmund Optics, Barrington, NJ) to ensure uniform field illumination. The emitted light is then directed through a custom FITC/GFP-optimized filter cube comprised of a shortpass filter (15-257, Edmund Optics), a dichroic mirror (DMLP490, Thorlabs), and a longpass filter (FGL495, Thorlabs). Furthermore, the open-architecture design explicitly supports the inclusion of auxiliary optical modules, allowing for the integration of additional components such as polarizers, phase plates, or spectral filters into the optical path, thereby enhancing the capabilities for advanced imaging modalities. The complete assembly integrates the LEGO frame, filter drawer, and precision gantry stage, as shown in Figure 1.

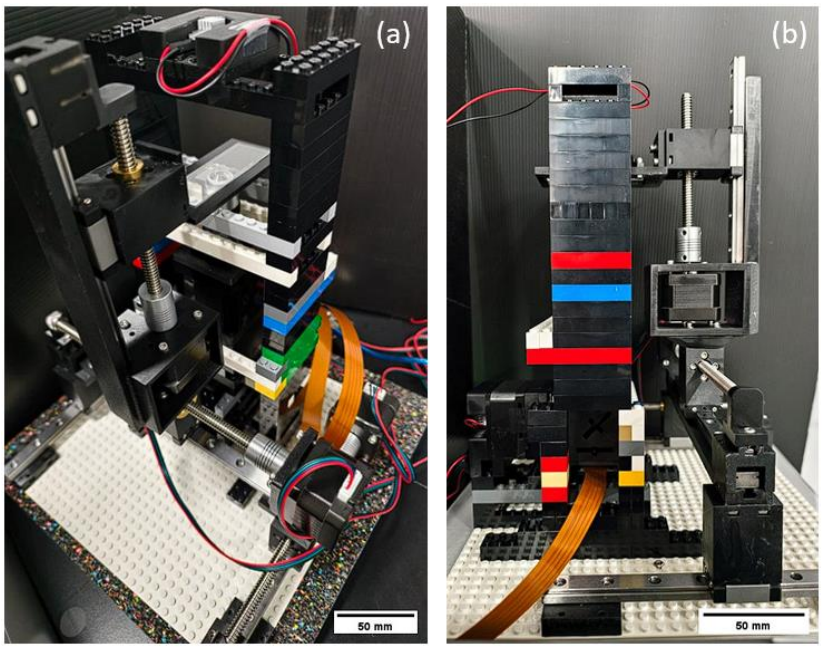

Figure 1. The fully assembled LEGO-based inverted fluorescence microscope system. The hybrid design combines a LEGO frame for structural adaptability with a rigid gantry stage for precision motion.

The sample stage employs a robust gantry architecture actuated by NEMA 17 stepper motors (17HS3401S, MotionKing, Changzhou, China). Linear motion along the X and Y axes is driven by 2-mm-pitch T8 lead screws with 1/16 microstepping, providing a theoretical positioning resolution of 0.625 µm. This configuration enables an extensive working volume of 70 × 50 × 70 mm, significantly surpassing that of flexure-based stages. Low-level motion control is implemented using an Arduino CNC Shield (Version 3.0, Protoneer, Wellington, New Zealand), which interfaces with a Raspberry Pi 5 (Raspberry Pi Ltd, Cambridge, UK) for synchronized command execution via USB.

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2.2. Software and Algorithms

2.2.1. Distributed Control System

The control software adopts a distributed architecture to ensure reliable operation during long-duration experiments. A Raspberry Pi hosts a custom Web-based graphical user interface (GUI), enabling users to access low-latency live video, perform coarse and fine stage positioning, and adjust exposure settings through any standard web browser. High-level GUI operations are decoupled from low-level motion control, which is executed by an Arduino CNC Shield, thereby preventing interface-induced delays from affecting critical autofocus routines. The GUI also includes a "One-Click Autofocus" function that initiates the Mountain-Climb Searching (MCS) algorithm [4] described in the next section.

2.2.2. Adaptive Autofocus Strategy

To enable robust unattended operation, the system employs a two-stage adaptive autofocus strategy based on a Mountain-Climb Searching (MCS) algorithm. The method dynamically switches focus metrics to accommodate brightfield and fluorescence imaging modes. During coarse search, global variance is used to rapidly identify sample features from a highly defocused state, producing a broad contrast landscape that prevents convergence to local maxima. Once near focus, the metric transitions to the Tenengrad operator—computed as the sum of squared Sobel gradients—to maximize sensitivity to high-frequency edge details and achieve precise focus locking.

For fluorescence imaging, where conventional metrics perform poorly due to low background contrast, a custom metric was developed. The fluorescence image is first processed with morphological opening to suppress diffuse haze, after which focus is evaluated using the mean gradient energy of the top 0.1% brightest pixels. This method selectively emphasizes true fluorescence structures while excluding dark background regions, enabling reliable autofocus under low signal-to-noise ratio (SNR) conditions.

2.2.3. Image Processing Pipeline

To compensate for vignetting and color crosstalk inherent to the Raspberry Pi Camera Module v2 (Raspberry Pi Ltd), an image-processing pipeline based on Bowman et al [3] was implemented. First, black-level subtraction was applied to the raw Bayer data to eliminate the fixed sensor pedestal and thermal noise, establishing a strictly linear baseline. Subsequently, flat-field correction using a Lens Shading Table (LST) was employed to rectify illumination falloff and ensure accurate white balance. Following brightfield calibration, the fluorescence modality requires two additional stages: background subtraction is first applied to attenuate stray light and enhance the signal-to-noise ratio, followed by a spatially varying color-correction matrix (CCM) to unmix the RGB channels and restore spectral fidelity.

2.3. Sample Preparation and Validation Targets

To validate the image processing pipeline and characterize the autofocus performance, two distinct reference targets were employed. First, a standard NBS 1952 Resolution Test Chart (NIST, Gaithersburg, MD) was imaged to verify the effectiveness of the flat-field correction and artifact removal algorithms. Following this, to prepare a biological reference sample for autofocus and fluorescence imaging validation, an aliquot of undiluted stock fluorescent microsphere suspension with a concentration of 1.0×10^6 beads/mL (FluoSpheres™ 15 µm yellow-green, Thermo Fisher Scientific, Waltham, MA) was transferred into a clean microcentrifuge tube. The suspension was thoroughly homogenized to ensure a uniform distribution of particles. From this mixture, 10 µL was withdrawn using a precision micropipette and deposited onto a clean standard glass microscope slide. Finally, a coverslip was gently placed over the droplet to create a thin, planar liquid layer, taking care to minimize the entrapment of air bubbles.

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3. EXPERIMENTAL RESULTS

3.1. Image Quality and Color Correction Verification

To validate the necessity of our image processing pipeline, we imaged a standard resolution test chart. The raw brightfield image (Figure 2(a)) exhibits significant vignetting and a strong green tint due to the specific Lens Shading Table (LST) characteristics of the Raspberry Pi Camera and its Bayer filter response. Furthermore, dust artifacts on the objective or sensor are clearly visible. By implementing LST and flat-field correction (Figure 2(b)), we successfully eliminated the vignetting and color cross-talk. Notably, this process also digitally removed the dust artifacts via the gain map, resulting in a clean and uniform background that preserves the high-frequency details of the resolution patterns.

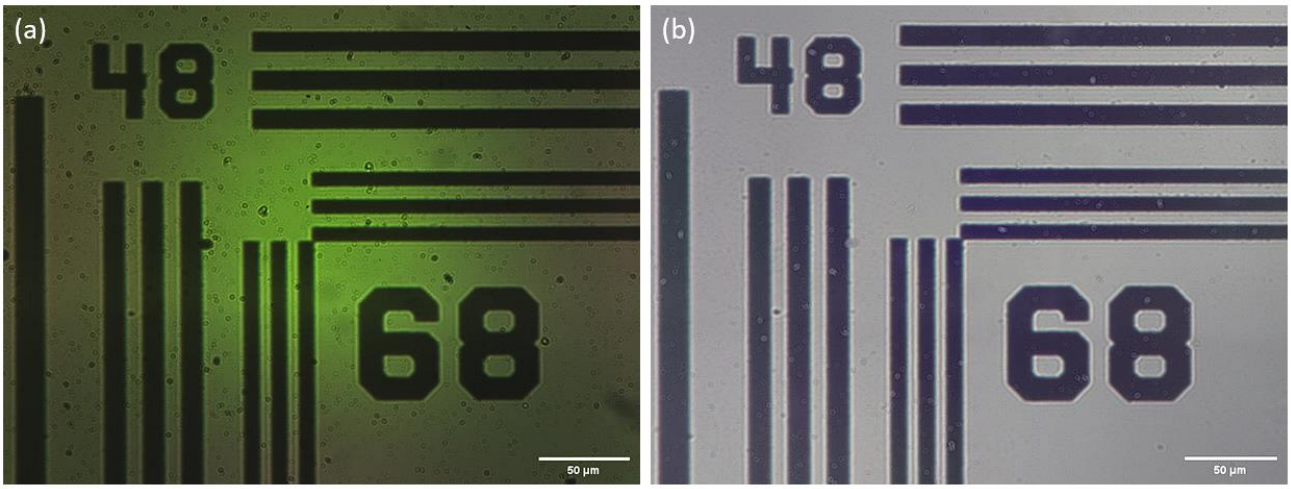

Figure 2. Validation of image correction using an NBS 1952 resolution test chart. (a) The raw brightfield image exhibiting significant vignetting, a characteristic green tint from the Bayer filter, and visible dust artifacts. (b) The result after Flat-field (LST) correction, demonstrating uniform illumination, a neutral background, and the effective removal of artifacts.

In fluorescence mode, raw sensor data often fail to capture the true spectral characteristics of the fluorophores. As illustrated in Figure 3(a), without the application of the CCM, the fluorescence signal suffers from low saturation, causing the green microspheres to appear washed out and whitish. This distortion arises from the sensor's broadband spectral sensitivity and crosstalk. By applying the calibrated CCM (Figure 3(b)), we successfully restored the spectral accuracy. The corrected image demonstrates high saturation and, crucially, achieves superior spatial color consistency, eliminating the color shift often observed between the center and edges of the fluorescent objects. Collectively, these correction steps transform the raw consumer-grade sensor data into research-grade imagery. By mitigating vignetting and spectral distortion, the system ensures that intensity values represent true biological signals rather than optical artifacts. This validation confirms that our calibrated computational pipeline successfully bridges the gap between low-cost modular hardware and the stringent requirements of quantitative biological imaging.

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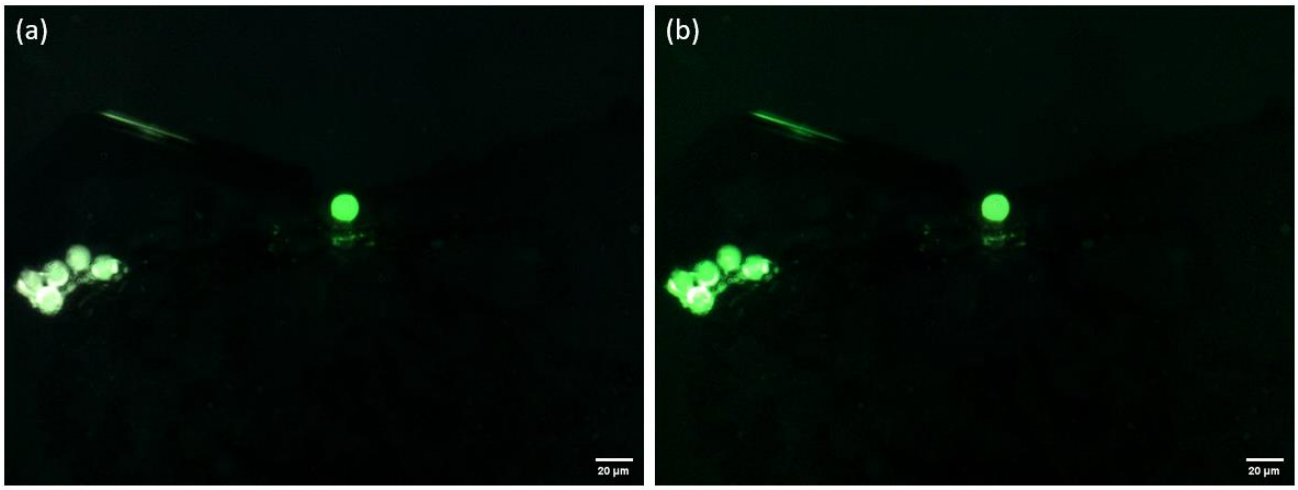

Figure 3. Impact of Color Correction Matrix (CCM) on fluorescence signal fidelity. (a) Image prior to CCM application. The signal exhibits low saturation and appears whitish (pale) due to the uncalibrated spectral response of the sensor. (b) The same image after applying the calibrated CCM. The correction restores the true vibrant green color of the fluorophores and ensures excellent chromatic uniformity, maintaining consistent color saturation from the center to the periphery of the microspheres.

3.2. Evaluation of Autofocus Performance

The efficacy of the adaptive autofocus algorithm was validated using Z-stacks of 15 µm FluoSpheres. As shown in Figure 4, the Global Variance metric (blue circle) exhibits a smooth gradient extending far from the focal plane, confirming its suitability for coarse searching. Conversely, the Tenengrad metric (red cross) produces a sharp, narrow peak, validating its use for fine-tuning. This hybrid approach achieved convergence over a 937.5 µm range within 14 seconds. This rapid and repeatable convergence serves as a direct functional validation of the rigid gantry architecture. The absence of focus oscillation or drift during the search process confirms that the LEGO-based chassis, reinforced with 3D-printed interfaces, maintains sufficient mechanical stiffness to support high-precision microstepping without the backlash often found in purely printed mechanisms.

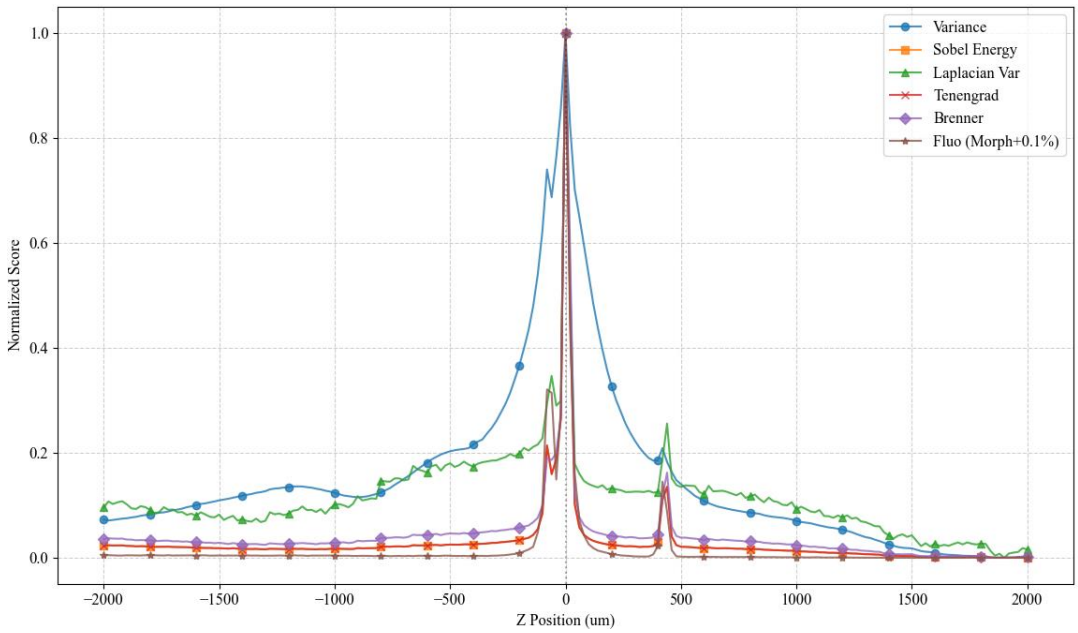

Figure 4. Comparison of normalized focus metric curves in brightfield mode. The Global Variance (blue) provides a broad search range, while Tenengrad (red) offers a sharp peak for precise locking.

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In fluorescence mode, background noise typically disrupts standard metrics. Figure 5 illustrates that while the Laplacian metric (green) suffers from a high noise baseline, our proposed "Morphological + Top 0.1%" metric (brown) effectively suppresses the haze, yielding a distinct, unimodal peak. This allowed the system to maintain the same 14-second convergence speed even in low-light conditions. These results demonstrate the robustness of the "Morphological + Top 0.1%" metric, proving that software algorithms can effectively compensate for the lower signal-to-noise ratio inherent in cost-effective sensors, thereby enabling reliable automation in non-ideal lighting environments.

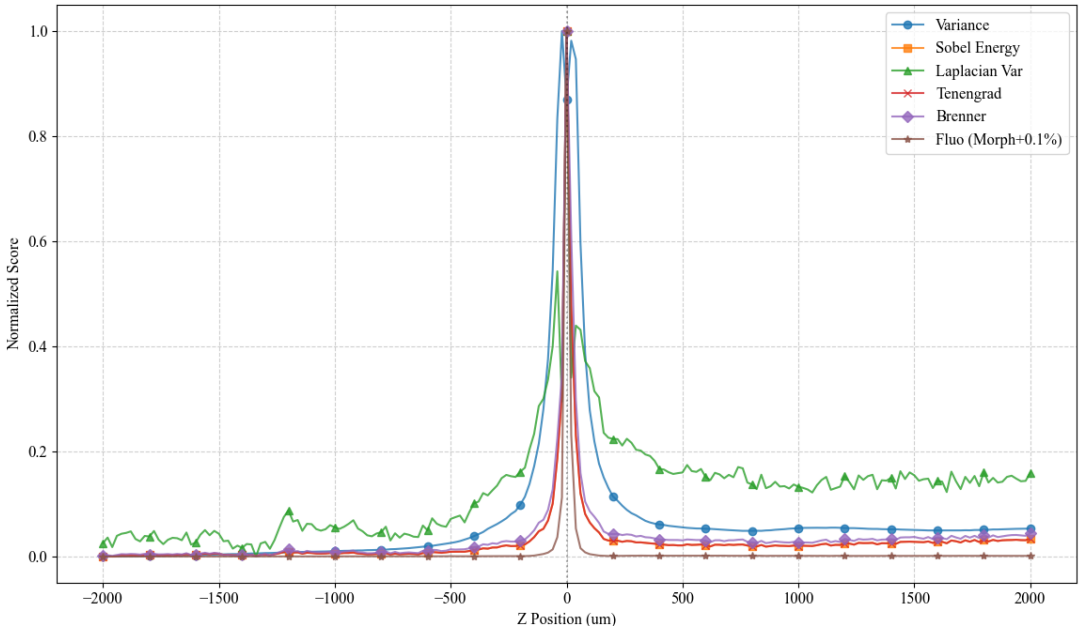

Figure 5. Evaluation of focus metrics in fluorescence mode. The proposed metric (brown star) eliminates the noise baseline observed in standard metrics, enabling reliable autofocus.

4. CONCLUSION

We have developed a cost-effective, fully automated inverted fluorescence microscope that achieves performance comparable to that of commercial systems in targeted applications. The experimental results validate the synergy between our specific hardware choices and algorithmic design. The integration of a rigid LEGO gantry mechanical design ensures the stability required for 0.625 µm positioning accuracy, while the two-stage “Coarse-to-Fine” autofocus algorithm enables reliable focus convergence within 14 seconds. Furthermore, the implementation of a calibrated color-correction pipeline successfully neutralizes the spectral limitations of consumer-grade sensors, ensuring high-fidelity data suitable for quantitative analysis.

Beyond its performance as a standalone instrument, the core innovation of this work lies in its "LEGO optical breadboard" philosophy. Unlike monolithic designs, the system features an open-architecture framework where the optical path and structural layout are fully user-definable. This high degree of modularity enables researchers to seamlessly interchange objective lenses, integrate custom optical modules, or reconfigure the light train without being constrained by manufacturing limitations. Consequently, the platform serves as a versatile, future-proof ecosystem. Future work will extend the platform to include multi-channel fluorescence capability and enhanced spectral separation, thereby broadening its applicability in agricultural phenotyping and biomedical monitoring.

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